

Ziyan_A8

Step0: Setup

modified the Labrole

LabRole

Info

Delete

Summary

Edit

Creation date

March 28, 2025, 11:48 (UTC-07:00)

ARN

arn:aws:iam::354754152263:role/LabRole

Instance profile ARN

arn:aws:iam::354754152263:instance-profile/LabInstanceProfile

Last activity

-

Maximum session duration

1 hour

Permissions

Trust relationships

Tags

(1)

Last Accessed

Revoke sessions

Permissions policies (3)

Info

Simulate

Remove

Add permissions

You can attach up to 10 managed policies.

Search

Filter by Type

All types

< 1 >

☐	Policy name	Type	Attached entities
☐	AdministratorAccess	AWS managed - job function	2
☐	AmazonEC2ContainerRegistryRead...	AWS managed	1
☐	AmazonSSMManagedInstanceCore	AWS managed	1

EC2 instance created with the modified LabRole attached

Instance summary for i-09a05693761bdd380 (ziyan) [Info](#)

Updated less than a minute ago

[Refresh](#)
[Connect](#)
[Instance state ▼](#)
[Actions ▼](#)

Instance ID i-09a05693761bdd380	Public IPv4 address 44.211.163.188 open address	Private IPv4 addresses 172.31.90.232
IPv6 address -	Instance state Running	Public IPv4 DNS ec2-44-211-163-188.compute-1.amazonaws.com open address
Hostname type IP name: ip-172-31-90-232.ec2.internal	Private IP DNS name (IPv4 only) ip-172-31-90-232.ec2.internal	Elastic IP addresses -
Answer private resource DNS name IPv4 (A)	Instance type t2.micro	AWS Compute Optimizer finding Opt-in to AWS Compute Optimizer for recommendations. Learn more
Auto-assigned IP address 44.211.163.188 [Public IP]	VPC ID vpc-04b616fed7889ef47	Auto Scaling Group name -
IAM Role LabRole	Subnet ID subnet-032772aac7abca751	Managed false
IMDSv2 Required	Instance ARN arn:aws:ec2:us-east-1:354754152263:instance/i-09a05693761bdd380	
Operator -		

Terraform and AWS CLI successfully installed on my instance

```

no VM guests are running out-of-the-box hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-90-232:~$ terraform -version
Terraform v1.11.3
on linux_amd64
ubuntu@ip-172-31-90-232:~$ aws --version
aws-cli/2.25.5 Python/3.12.9 Linux/6.8.0-1024-aws exe/x86_64.ubuntu.24
ubuntu@ip-172-31-90-232:~$
  
```

Step1

VPC create

Screenshot: Capture the VPC creation summary page showing the VPC ID, name, and CIDR block (Screenshot #1).

ie999a0b93264

vpc-0fa26e999a0b93264 / webapp-vpc [Actions](#)

Details [Info](#)

VPC ID  vpc-0fa26e999a0b93264	State  Available	Block Public Access  Off	DNS hostnames Enabled
DNS resolution Enabled	Tenancy default	DHCP option set dopt-0c8ac7e1d401f0ec7	Main route table rtb-0e3bc71b371c2b04e
Main network ACL acl-0e5f833f1ac344c44	Default VPC No	IPv4 CIDR 10.0.0.0/16	IPv6 pool -
IPv6 CIDR (Network border group) -	Network Address Usage metrics Disabled	Route 53 Resolver DNS Firewall rule groups -	Owner ID  354754152263

[Resource map](#) | [CIDRs](#) | [Flow logs](#) | [Tags](#) | [Integrations](#)

Resource map [Info](#)

VPC [Show details](#)
Your AWS virtual network

webapp-vpc

Subnets (0)
Subnets within this VPC

Route tables (1)
Route network traffic to resources

rtb-0e3bc71b371c2b04e

Screenshot: Capture subnet configurations, including CIDR blocks and availability zones (Screenshot #2)

Public subnet 1

subnet-03934d06b5f341534 / Public Subnet 1

[Actions](#)

Details

Subnet ID

 subnet-03934d06b5f341534

IPv4 CIDR

 10.0.1.0/24

Availability Zone

 us-east-1a

Route table

 rtb-08b39f61c81c356dc | [Public Route Table](#)

Auto-assign IPv6 address

No

IPv4 CIDR reservations

–

Resource name DNS A record

Disabled

Subnet ARN

 arn:aws:ec2:us-east-1:354754152263:subnet/subnet-03934d06b5f341534

Available IPv4 addresses

 251

Availability Zone ID

 use1-az2

Network ACL

 acl-0e5f833f1ac344c44

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

–

Resource name DNS AAAA record

Disabled

State

 Available

IPv6 CIDR

–

Network border group

 us-east-1

Default subnet

No

Customer-owned IPv4 pool

–

IPv6-only

No

DNS64

Disabled

Block Public Access

 Off

IPv6 CIDR association ID

–

VPC

 vpc-0fa26e999a0b93264 | [webapp-vpc](#)

Auto-assign public IPv4 address

Yes

Outpost ID

–

Hostname type

IP name

Owner

 354754152263

Public subnet 2

subnet-0b01b454b3d5e4e0a / Public Subnet 2

[Actions](#)

Details

Subnet ID

 subnet-0b01b454b3d5e4e0a

IPv4 CIDR

 10.0.2.0/24

Availability Zone

 us-east-1b

Route table

 rtb-08b39f61c81c356dc | [Public Route Table](#)

Auto-assign IPv6 address

No

IPv4 CIDR reservations

–

Resource name DNS A record

Disabled

Subnet ARN

 arn:aws:ec2:us-east-1:354754152263:subnet/subnet-0b01b454b3d5e4e0a

Available IPv4 addresses

 251

Availability Zone ID

 use1-az4

Network ACL

 acl-0e5f833f1ac344c44

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

–

Resource name DNS AAAA record

Disabled

State

 Available

IPv6 CIDR

–

Network border group

 us-east-1

Default subnet

No

Customer-owned IPv4 pool

–

IPv6-only

No

DNS64

Disabled

Block Public Access

 Off

IPv6 CIDR association ID

–

VPC

 vpc-0fa26e999a0b93264 | [webapp-vpc](#)

Auto-assign public IPv4 address

Yes

Outpost ID

–

Hostname type

IP name

Owner

 354754152263

Public subnet 3

subnet-00f3b8f321e425311 / Public Subnet 3

Actions

Details			
Subnet ID subnet-00f3b8f321e425311	Subnet ARN arn:aws:ec2:us-east-1:354754152263:subnet/subnet-00f3b8f321e425311	State Available	Block Public Access Off
IPv4 CIDR 10.0.3.0/24	Available IPv4 addresses 251	IPv6 CIDR -	IPv6 CIDR association ID -
Availability Zone us-east-1c	Availability Zone ID use1-az6	Network border group us-east-1	VPC vpc-0fa26e999a0b93264 webapp-vpc
Route table rtb-08b39f61c81c356dc Public Route Table	Network ACL acl-0e5f833f1ac344c44	Default subnet No	Auto-assign public IPv4 address Yes
Auto-assign IPv6 address No	Auto-assign customer-owned IPv4 address No	Customer-owned IPv4 pool -	Outpost ID -
IPv4 CIDR reservations -	IPv6 CIDR reservations -	IPv6-only No	Hostname type IP name
Resource name DNS A record Disabled	Resource name DNS AAAA record Disabled	DNS64 Disabled	Owner 354754152263

private subnet1

subnet-0ce6712bf17cef180 / Private Subnet 1

Actions

Details			
Subnet ID subnet-0ce6712bf17cef180	Subnet ARN arn:aws:ec2:us-east-1:354754152263:subnet/subnet-0ce6712bf17cef180	State Available	Block Public Access Off
IPv4 CIDR 10.0.4.0/24	Available IPv4 addresses 251	IPv6 CIDR -	IPv6 CIDR association ID -
Availability Zone us-east-1a	Availability Zone ID use1-az2	Network border group us-east-1	VPC vpc-0fa26e999a0b93264 webapp-vpc
Route table rtb-0a44af03be5a5df50 Private Route Table	Network ACL acl-0e5f833f1ac344c44	Default subnet No	Auto-assign public IPv4 address No
Auto-assign IPv6 address No	Auto-assign customer-owned IPv4 address No	Customer-owned IPv4 pool -	Outpost ID -
IPv4 CIDR reservations -	IPv6 CIDR reservations -	IPv6-only No	Hostname type IP name
Resource name DNS A record Disabled	Resource name DNS AAAA record Disabled	DNS64 Disabled	Owner 354754152263

private subnet 2

subnet-014746d471d88f178 / Private Subnet 2

Actions

Details

Subnet ID

subnet-014746d471d88f178

IPv4 CIDR

10.0.5.0/24

Availability Zone

us-east-1b

Route table

rtb-0a44af03be5a5df50 | Private Route Table

Auto-assign IPv6 address

No

IPv4 CIDR reservations

-

Resource name DNS A record

Disabled

Subnet ARN

arn:aws:ec2:us-east-1:354754152263:subnet/subnet-014746d471d88f178

Available IPv4 addresses

251

Availability Zone ID

use1-az4

Network ACL

acl-0e5f833f1ac344c44

Auto-assign customer-owned IPv4 address

No

IPv6 CIDR reservations

-

Resource name DNS AAAA record

Disabled

State

Available

IPv6 CIDR

-

Network border group

us-east-1

Default subnet

No

Customer-owned IPv4 pool

-

IPv6-only

No

DNS64

Disabled

Block Public Access

Off

IPv6 CIDR association ID

-

VPC

vpc-0fa26e999a0b93264 | webapp-vpc

Auto-assign public IPv4 address

No

Outpost ID

-

Hostname type

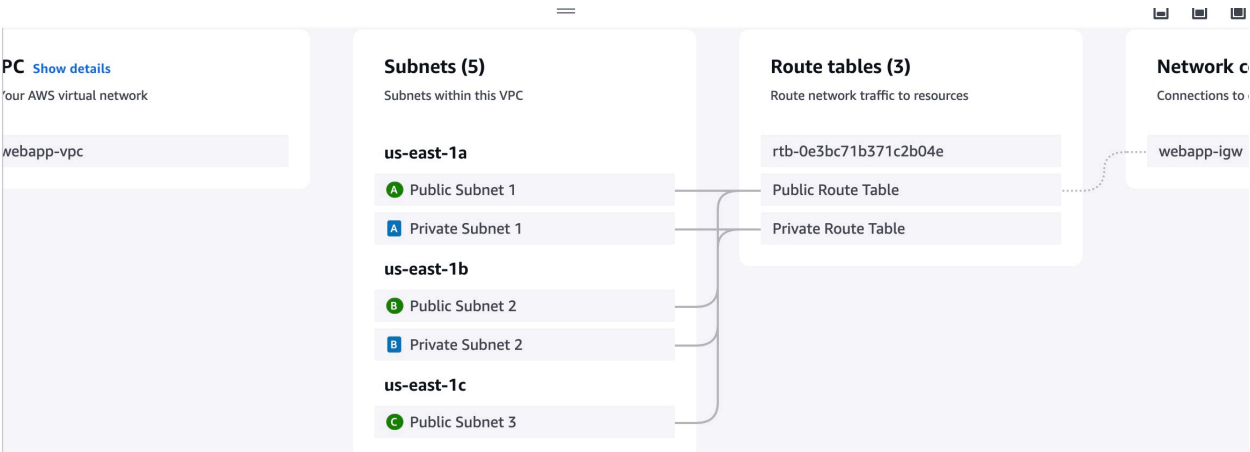
IP name

Owner

354754152263

Screenshot: Show IGW attachment and route table configurations (Screenshot #3)

Resource map to show the logic under VPC is correct



IGW

igw-02776d2157b49f6f5 / webapp-igw

Actions

Details Info

Internet gateway ID

igw-02776d2157b49f6f5

State

Attached

VPC ID

vpc-0fa26e999a0b93264 | webapp-vpc

Owner

354754152263

Tags

Manage tags

Search tags

< 1 > ⚙

Key	Value
Name	webapp-igw

Public route table

rtb-08b39f61c81c356dc / Public Route Table

Actions

Details Info

Route table ID

rtb-08b39f61c81c356dc

Main

No

Explicit subnet associations

3 subnets

Edge associations

-

VPC

vpc-0fa26e999a0b93264 | webapp-vpc

Owner ID

354754152263

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (2)

Both Edit routes

Filter routes

< 1 > ⚙

Destination	Target	Status	Propagated
0.0.0.0/0	igw-02776d2157b49f6f5	Active	No
10.0.0.0/16	local	Active	No

private route table

rtb-0a44af03be5a5df50 / Private Route Table

Actions

Details

Info

Route table ID

rtb-0a44af03be5a5df50

VPC

vpc-0fa26e999a0b93264 | webapp-vpc

Main

No

Owner ID

354754152263

Explicit subnet associations

2 subnets

Edge associations

-

Routes

Subnet associations

Edge associations

Route propagation

Tags

Routes (1)

Both

Edit routes

Filter routes

1

Settings

Destination	Target	Status	Propagated
10.0.0.0/16	local	Active	No

Step2

Screenshot: Capture ALB security group rules for both inbound and outbound traffic (Screenshot #4).

inbound - Allow HTTP traffic on port 80 from anywhere (0.0.0.0/0)

sg-089755b2132f15244 - alb-sg

Actions

Details

Security group name

alb-sg

Security group ID

sg-089755b2132f15244

Description

Allow HTTP from anywhere

VPC ID

vpc-0fa26e999a0b93264

Owner

354754152263

Inbound rules count

1 Permission entry

Outbound rules count

1 Permission entry

Inbound rules

Outbound rules

Sharing - new

VPC associations - new

Tags

Inbound rules (1)

Manage tags

Edit inbound rules

Search

1

Settings

Security group rule ID	IP version	Type	Protocol	Port range	Source
sgr-0a33edf2c86911dc8	IPv4	HTTP	TCP	80	0.0.0.0/0

Outbound -Allow all traffic (0.0.0.0/0)

sg-089755b2132f15244 - alb-sg

[Actions](#)

Details

Security group name
alb-sg

Security group ID
sg-089755b2132f15244

Description
Allow HTTP from anywhere

VPC ID
vpc-0fa26e999a0b93264

Owner
354754152263

Inbound rules count
1 Permission entry

Outbound rules count
1 Permission entry

[Inbound rules](#)[Outbound rules](#)[Sharing - new](#)[VPC associations - new](#)[Tags](#)

Outbound rules (1)

[Manage tags](#)[Edit outbound rules](#)[< 1 >](#)

Security group rule ID	IP version	Type	Protocol	Port range	Destination
sgr-01dedd0761ca4cbf9	IPv4	All traffic	All	All	0.0.0.0/0

Screenshot: Show EC2 instances' security group settings (Screenshot #5)

Inbound attached the ALB-SG source

sg-0921cecc1487507bc - ec2-sg

[Actions](#)

Details

Security group name
ec2-sg

Security group ID
sg-0921cecc1487507bc

Description
Allow HTTP from ALB

VPC ID
vpc-0fa26e999a0b93264

Owner
354754152263

Inbound rules count
1 Permission entry

Outbound rules count
1 Permission entry

[Inbound rules](#)[Outbound rules](#)[Sharing - new](#)[VPC associations - new](#)[Tags](#)

Inbound rules (1)

[Manage tags](#)[Edit inbound rules](#)[< 1 >](#)

Security group rule ID	IP version	Type	Protocol	Port range	Source
sgr-097da3569286ff64c	-	Custom TCP	TCP	8080	sg-089755b2132f15244

Outbound: Allow all traffic (0.0.0.0/0).

sg-0921cecc1487507bc - ec2-sg

Actions ▾

Details

Security group name
ec2-sg

Security group ID
sg-0921cecc1487507bc

Description
Allow HTTP from ALB

VPC ID
vpc-0fa26e999a0b93264

Owner
354754152263

Inbound rules count
1 Permission entry

Outbound rules count
1 Permission entry

Inbound rules **Outbound rules** Sharing - new VPC associations - new Tags

Outbound rules (1)



Manage tags

Edit outbound rules

Search

< 1 > ⚙️

Security group rule ID ▾	IP version ▾	Type ▾	Protocol ▾	Port range ▾	Destination
sgr-0e0b4926ac87c55d4	IPv4	All traffic	All	All	0.0.0.0/0

Screenshot: Show RDS security group settings (Screenshot #6)

Inbound - chose the MySQL/Aurora type, and **WebApp-SG** (security group ID) as the source

sg-0f1b4f4050ee36cea - rds-sg

Actions ▾

Details

Security group name
rds-sg

Security group ID
sg-0f1b4f4050ee36cea

Description
Allow MySQL from EC2 SG

VPC ID
vpc-0fa26e999a0b93264

Owner
354754152263

Inbound rules count
1 Permission entry

Outbound rules count
1 Permission entry

Inbound rules Outbound rules Sharing - new VPC associations - new Tags

Inbound rules (1)



Manage tags

Edit inbound rules

Search

< 1 > ⚙️

Security group rule ID ▾	IP version ▾	Type ▾	Protocol ▾	Port range ▾	Source
sg-55c655f14223d1da	-	MySQL/Aurora	TCP	3306	sg-0921cecc1487507bc

Outbound:

sg-0f1b4f4050ee36cea - rds-sg

[Actions ▾](#)

Details

Security group name
rds-sg

Security group ID
sg-0f1b4f4050ee36cea

Description
Allow MySQL from EC2 SG

VPC ID
[vpc-0fa26e999a0b93264](#)

Owner
354754152263

Inbound rules count
1 Permission entry

Outbound rules count
1 Permission entry

[Inbound rules](#)[Outbound rules](#)[Sharing - new](#)[VPC associations - new](#)[Tags](#)

Outbound rules (1)

[Manage tags](#)[Edit outbound rules](#)[< 1 > ⚙](#)

Security group rule ID ▾	IP version ▾	Type ▾	Protocol ▾	Port range ▾	Destination
sgr-06934500c2bd88491	IPv4	All traffic	All	All	0.0.0.0/0

Step3

in terminal, the RDS created successfully

```
aws_db_instance.webapp_db: Still creating... [3m10s elapsed]
aws_db_instance.webapp_db: Still creating... [3m20s elapsed]
aws_db_instance.webapp_db: Still creating... [3m30s elapsed]
aws_db_instance.webapp_db: Still creating... [3m40s elapsed]
aws_db_instance.webapp_db: Still creating... [3m50s elapsed]
aws_db_instance.webapp_db: Still creating... [4m0s elapsed]
aws_db_instance.webapp_db: Still creating... [4m10s elapsed]
aws_db_instance.webapp_db: Still creating... [4m20s elapsed]
aws_db_instance.webapp_db: Still creating... [4m30s elapsed]
aws_db_instance.webapp_db: Still creating... [4m40s elapsed]
aws_db_instance.webapp_db: Still creating... [4m50s elapsed]
aws_db_instance.webapp_db: Still creating... [5m0s elapsed]
aws_db_instance.webapp_db: Still creating... [5m10s elapsed]
aws_db_instance.webapp_db: Creation complete after 5m14s [id=db-6ICATV3IKIKGI
LSWY6RKSBM]
```

Apply complete! Resources: 20 added, 0 changed, 0 destroyed.

ubuntu@ip-172-31-90-232:~\$

Screenshot: RDS instance configuration summary, showing instance class, engine, and network settings (Screenshot #7)

In dashboard, double checked the RDS created successfully

Instance class: t3.micro

Engine: PostgreSQL

network setting: subnets - two private subnets

webapp-db

webapp-db

Summary

DB identifier webapp-db	Status Available	Role Instance	Engine PostgreSQL	Recommendations
CPU -	Class db.t3.micro	Current activity	Region & AZ us-east-1a	

Connectivity & security | Monitoring | Logs & events | Configuration | Maintenance & backups | Data migrations - new

Connectivity & security

Endpoint & port	Networking	Security
Endpoint webapp-db.cjkua6kawr0j.us-east-1.rds.amazonaws.com	Availability Zone us-east-1a	VPC security groups rds-sg (sg-00ca60ff36f117a94) Active
Port 5432	VPC webapp-vpc (vpc-0f00bb978194eacd3)	Publicly accessible No
	Subnet group webapp-db-subnet-group	Certificate authority Info rds-ca-rsa2048-g1
	Subnets subnet-026f2035ef8d5545f subnet-026eb3e363e1dd755	Certificate authority date May 25, 2061, 16:34 (UTC-07:00)
		DB instance certificate expiration

Step4

Screenshot: Launch Template configuration page, including the instance type and user data script (Screenshot #8).

set successfully with provided AMI ID

webapp-lt-20250328204059939200000001 (lt-0cfe7be313e1cf962)

Actions ▾

Delete template

Launch template details

Launch template ID
lt-0cfe7be313e1cf962

Launch template name
webapp-lt-20250328204059939200000001

Default version
1

Owner
arn:aws:sts::354754152263:assumed-role/LabRole/i-09a05693761bdd380

Details

Versions

Template tags

Launch template version details

Actions ▾

Delete template version

Version

1 (Default) ▾

Description

-

Date created

2025-03-28T20:41:00.000Z

Created by

arn:aws:sts::354754152263:assumed-role/LabRole/i-09a05693761bdd380

Instance details

Storage

Resource tags

Network interfaces

Advanced details

AMI ID

ami-03a6c16a66bbe0a7a

Instance type

t3.micro

Availability Zone

-

Key pair name

-

Security groups

-

Security group IDs

sg-0a569e812720e269e

User data as follows:

webapp-lt-20250328204059939200000001

-	-	-
Token hop limit	Metadata version	Metadata version
-	-	-
Instance Configurable Bandwidth		
-		
User data		
		
<pre>#!/bin/bash # Create .env file with DB connection info from Terraform variables cat <<EOT > /usr/bin/csye6225/.env DB_HOST=webapp-db.cjkua6kawr0j.us-east-1.rds.amazonaws.com DB_PORT=3306 DB_USERNAME=webappadmin DB_PASSWORD=password789! DB_NAME=webapp_db EOT # Set secure file permissions sudo chmod 600 /usr/bin/csye6225/.env sudo chown csye6225:csye6225 /usr/bin/csye6225/.env # Start the webapp service sudo systemctl start webapp.service</pre>		
Base64-encoded user data has been decoded for readability.		

step5

Screenshot: Target group settings, including health check configurations
(Screenshot #9)

Load balancer created

▼ Details

Load balancer type

Application

Scheme

Internet-facing

Status

✔ Active

Hosted zone

Z35SXDOTRQ7X7K

VPC

[vpc-0c2b2b9c9f49a1563](#)

Availability Zones

[subnet-0549ca7fdd62e38c2](#) us-east-1c (use1-az6)
[subnet-013a136c74458b8d4](#) us-east-1a (use1-az2)
[subnet-082f462834dd7c8be](#) us-east-1b (use1-az4)

Load balancer IP address type

IPv4

Date created

March 28, 2025, 13:46 (UTC-07:00)

Load balancer ARN

[arn:aws:elasticloadbalancing:us-east-1:354754152263:loadbalancer/app/webapp-alb/20825a47a21f998a](#)

DNS name

[webapp-alb-2111293451.us-east-1.elb.amazonaws.com](#) (A Record)

< Load rules

Network mapping

Resource map

Security

Monitoring

Integrations

Attributes

Capacity

Tags

>

Listeners and rules (1) [Info](#)

🔄

Manage rules ▼

Manage listener ▼

Add listener

A listener checks for connection requests on its configured protocol and port. Traffic received by the listener is routed according to the default action and any additional rules.

🔍 Filter listeners

< 1 > ⚙️

☐	Protocol:Port ▼	Default action ▼	Rules ▼	ARN ▼	Security policy ▼	Default
☐	HTTP:80	Forward to target group webapp-target-group: 1 (100%) - Target group stickiness: Off	1 rule	ARN	Not applicable	Not app

Now it is empty for the target because the ASG haven't been created (see target group and passed health check in next step after the ASG created)

webapp-target-group

[Actions](#)

Details

 [arn:aws:elasticloadbalancing:us-east-1:354754152263:targetgroup/webapp-target-group/b65a5eea82972b82](#)

Target type

Instance

Protocol : Port

HTTP: 8080

Protocol version

HTTP1

VPC

[vpc-0f00bb978194eacd3](#)

IP address type

IPv4

Load balancer

[webapp-alb](#)

0

Total targets

 0

Healthy

0 Anomalous

 0

Unhealthy

 0

Unused

 0

Initial

 0

Draining

[Targets](#)

[Monitoring](#)

[Health checks](#)

[Attributes](#)

[Tags](#)

Registered targets (0) [Info](#)

 [Anomaly mitigation: Not applicable](#)



[Deregister](#)

[Register targets](#)

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.



< 1 > 

☐	Instance ID	Name	Port	Zone	Health status	Health status details	Admini...
--------------------------	-------------	------	------	------	---------------	-----------------------	-----------

No registered targets

Health check setting

[Targets](#)

[Monitoring](#)

[Health checks](#)

[Attributes](#)

[Tags](#)

Health check settings

[Edit](#)

Protocol

HTTP

Path

/healthz

Port

Traffic port

Healthy threshold

2 consecutive health check successes

Unhealthy threshold

2 consecutive health check failures

Timeout

5 seconds

Interval

30 seconds

Success codes

200

After the ASG is created in step6

Health check in Console: 1 healthy target registered

webapp-target-group

[Actions](#)

Details

[arn:aws:elasticloadbalancing:us-east-1:354754152263:targetgroup/webapp-target-group/2b2b0d5090c437c2](#)

Target type
Instance

Protocol : Port
HTTP: 8080

Protocol version
HTTP1

VPC
[vpc-0c2b2b9c9f49a1563](#)

IP address type
IPv4

Load balancer
[webapp-alb](#)

1

Total targets

✓ 1

Healthy

0 Anomalous

✗ 0

Unhealthy

⏸ 0

Unused

⏸ 0

Initial

⏸ 0

Draining

TARGETS

Monitoring

Health checks

Attributes

Tags

Registered targets (1) [Info](#)

Anomaly mitigation: **Not applicable**

[Deregister](#)

[Register targets](#)

Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.

1

☐	Instance ID	Name	Port	Zone	Health status	Health status details	Admini...
--------------------------	-------------	------	------	------	---------------	-----------------------	-----------

step6

Screenshot: ASG settings, including instance counts (Screenshot #10).

webapp-asg

webapp-asg Capacity overview

[Edit](#)

[arn:aws:autoscaling:us-east-1:354754152263:autoScalingGroup:7dffd9e-a9f1-4ba4-a226-88b82400cdde:autoScalingGroupName/webapp-asg](#)

Desired capacity
1

Scaling limits (Min - Max)
1 - 3

Desired capacity type
Units (number of instances)

Status
-

Date created

Fri Mar 28 2025 14:00:27 GMT-0700 (Pacific Daylight Time)

Details

Integrations - new

Automatic scaling

Instance management

Instance refresh

Activity

Monitoring

Launch template

[Edit](#)

Launch template

[lt-0cfe7be313e1cf962](#)
webapp-lt-
20250328204059939200000001

AMI ID
 [ami-03a6c16a66bbe0a7a](#)

Instance type
t3.micro

Owner
[arn:aws:sts::354754152263:assumed-role/LabRole/i-09a05693761bdd380](#)

Version
Latest

Security groups
-

Security group IDs
 [sg-0a569e812720e269e](#)

Create time
Fri Mar 28 2025 13:41:00 GMT-0700
(Pacific Daylight Time)

Description
-

Storage (volumes)
-

Key pair name
-

Request Spot Instances
No

[View details in the launch template console](#)

instance counts

webapp-asg ⓘ 🔗

webapp-asg Capacity overview Edit

[arn:aws:autoscaling:us-east-1:354754152263:autoScalingGroup:7dffd9e-a9f1-4ba4-a226-88b82400cdde:autoScalingGroupName/webapp-asg](#)

Desired capacity 1	Scaling limits (Min - Max) 1 - 3	Desired capacity type Units (number of instances)	Status -
------------------------------	--	---	--------------------

Date created
Fri Mar 28 2025 14:00:27 GMT-0700 (Pacific Daylight Time)

[Details](#) | [Integrations - new](#) | [Automatic scaling](#) | [Instance management](#) | [Instance refresh](#) | [Activity](#) | [Monitoring](#)

Instances (3) 🔄 Actions

☐	Instance ID	Lifecycle	Instanc...	Weight...	Launch...	Availab...	Health ...	Protecte
☐	i-035215bbcf5ed74f6	Terminating	t3.micro	-	webapp-lt-202!	us-east-1a	⚠️ Unhealthy..	
☐	i-046470c92f1f9cf5e	InService	t3.micro	-	webapp-lt-202!	us-east-1a	✅ Healthy	
☐	i-0bce7a29819a1f5b5	Terminating	t3.micro	-	webapp-lt-202!	us-east-1a	⚠️ Unhealthy..	

CloudWatch alarm configurations for scaling policies (Screenshot #11)

In CloudWatch, we can see LowCPU is in alarm, and HighCPU is ok

Alarms (2) ☐ Hide Auto Scaling alarms Clear selection 🔄 Create composite alarm Actions Create alarm

Alarm state: Any Alarm type: Any Actions status: Any < 1 > ⚙️

☐	Name	State	Last state update (UTC)	Conditions	Actions
☐	LowCPUUtilization	⚠️ In alarm	2025-03-28 21:02:42	CPUUtilization < 30 for 2 datapoints within 2 minutes	✅ Actions enab
☐	HighCPUUtilization	✅ OK	2025-03-28 21:02:33	CPUUtilization > 70 for 2 datapoints within 2 minutes	✅ Actions enab

step7

Screenshot: Scaling up and down of instances in the ASG during load testing (Screenshot #12)

scale up with 4 instances (prior is 3)

Instances (4)

Filter instances

☐	Instance ID	Lifecycle	Instanc...	Weight...	Launch...	Availab...	Health ...	Protecte..
☐	i-033d06ab73af424df	InService	t3.micro	-	webapp-lt-202!	us-east-1a	Healthy	
☐	i-059f0bc94c6e04b37	Terminating	t3.micro	-	webapp-lt-202!	us-east-1a	Unhealthy..	
☐	i-0bb4642761325a955	Terminating	t3.micro	-	webapp-lt-202!	us-east-1a	Unhealthy..	
☐	i-0c9ac6a6a8e3541b9	Terminating	t3.micro	-	webapp-lt-202!	us-east-1a	Unhealthy..	

Increase the load test intensity by raising the total number of requests and concurrency level (e.g., 500 concurrent users × 50,000 requests).

We can see the instances scale up to 5

webapp-asg Capacity overview [Edit](#)

[arn:aws:autoscaling:us-east-1:354754152263:autoScalingGroup:7dffd9e-a9f1-4ba4-a226-88b82400cdd:autoScalingGroupName/webapp-asg](#)

Desired capacity 1	Scaling limits (Min - Max) 1 - 3	Desired capacity type Units (number of instances)	Status Updating capacity
------------------------------	--	---	------------------------------------

Date created
Fri Mar 28 2025 14:00:27 GMT-0700 (Pacific Daylight Time)

[Details](#) | [Integrations - new](#) | [Automatic scaling](#) | [Instance management](#) | [Instance refresh](#) | [Activity](#) | [Monitoring](#)

Instances (5)

Filter instances

☐	Instance ID	Lifecycle	Instanc...	Weight...	Launch...	Availab...	Health ...	Protecte..
☐	i-002f0e4c1f7c0fca0	Pending	t3.micro	-	webapp-lt-202!	us-east-1a	Healthy	
☐	i-01c1ab5c6f3bbc313	Terminating	t3.micro	-	webapp-lt-202!	us-east-1a	Unhealthy..	
☐	i-033d06ab73af424df	Terminating	t3.micro	-	webapp-lt-202!	us-east-1a	Unhealthy..	
☐	i-0bb4642761325a955	Terminating	t3.micro	-	webapp-lt-202!	us-east-1a	Unhealthy..	
☐	i-0c9ac6a6a8e3541b9	Terminating	t3.micro	-	webapp-lt-202!	us-east-1a	Unhealthy..	

This screenshot shows the Auto Scaling Group (ASG) automatically scaling down after the load test. The desired capacity has returned to 1 instance, and three EC2 instances are in the "Terminating" state. Only one instance remains "InService"

and marked "Healthy", indicating that the ASG successfully reduced resources based on decreased CPU utilization.

After performing a high-concurrency load test, the ASG triggered a scale-up and then scaled back down as the load decreased.

webapp-asg

webapp-asg Capacity overview [Edit](#)

arn:aws:autoscaling:us-east-1:354754152263:autoScalingGroup:7dffd9e-a9f1-4ba4-a226-88b82400cde:autoScalingGroupName/webapp-asg

Desired capacity 1	Scaling limits (Min - Max) 1 - 3	Desired capacity type Units (number of instances)	Status Updating capacity
------------------------------	--	---	------------------------------------

Date created
Fri Mar 28 2025 14:00:27 GMT-0700 (Pacific Daylight Time)

Details | Integrations - new | Automatic scaling | **Instance management** | Instance refresh | Activity | Monitoring

Instances (4) [Actions](#)

Filter instances

☐	Instance ID	Lifecycle	Instanc...	Weight...	Launch...	Availab...	Health ...	Protecte...
☐	i-002f0e4c1f7c0fca0	Terminating	t3.micro	-	webapp-lt-202!	us-east-1a	Unhealthy..	
☐	i-01c1ab5c6f3bbc313	Terminating	t3.micro	-	webapp-lt-202!	us-east-1a	Unhealthy..	
☐	i-098c5c2396eb11fd6	Terminating	t3.micro	-	webapp-lt-202!	us-east-1a	Unhealthy..	
☐	i-0db270c5eed92507e	InService	t3.micro	-	webapp-lt-202!	us-east-1a	Healthy	

Verify Application Functionality

a. Test Create User and Get User APIs.

b. Screenshot: Request and response for each API call (Screenshot #13)

In Postman, Create User

After send, I got 201 created, means POST /v1/user is passed

POST / Create User Save

3T ▼ http://webapp-alb-203047056.us-east-1.elb.amazonaws.com/v1/user Send

Headers (8) Body Params Cache Pre-req. Settings Cookie

Content Type: ☐ form-urlencoded ☐ raw ☐ binary ☐ GraphQL ☒ JSON

```
1 {"email": "sample.email@example.com",
2  "password": "samplePassword123",
3  "first_name": "User",
4  "last_name": "User"}
```

Cookies Headers (6) Test Results 201 Created Time: 76 ms Save Response

Body Cookies Headers Visualize JSON Copy

```
{
  "id": "760d012de-d878-4017-b1f6-19584787800",
  "email": "sample.email@example.com",
  "first_name": "Sample",
  "account_created": "Sample",
  "account_updated": "2025-4-09T19:28.370Z"
}
```

Get User Details, with 200 ok status and basic Auth with the created user's username and password

SYN 6225 / Get User Details GET http://webapp-alb-203047056.us-east-1.elb.amazonaws.com/v1/user/self Send

Params Authorization Headers (9) Body Pre-request Script Tests Settings Cookies

Type Basic Auth Basic Auth sample.email@example.com Username Show Password

Body Cookies Headers (6) Test Results Status: 200 OK Time: 198 ms Size: 414 B

Pretty Raw Preview V/Z.A JSON

```
1 {
2   "id": "7b607130:66f2-4b96-a4a3-c28af9d70ea",
3   "email": "sample.email@example.com",
4   "first_name": "Sample",
5   "last_name": "User",
6   "account_created": "2025-04-09T14:15:09Z",
7   "updated_at": "2025-04-09T14:15:09Z"
8 }
```

Health Check

HTTP

CSYE 6225 / Health Check

Save

Share

GET

http://webapp-alb-203047056.us-east-1.elb.amazonaws.com/healthz

Send

Params

Authorization

Headers (6)

Body

Scripts

Settings

Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body

Cookies (5)

Headers (16)

Test Results

200 OK

1166 ms

618 B

Save Response

Pretty

Raw

Preview

Visualize

JSON

1

2

3

4

"status": "OK",

"version": "v3"